

Trido Model Card: trido-model:latest

Flagship Spatial AI Smartboard Co-Worker • Local-First, Zero-Login, Continuous In-Place Mutation

BASE MODEL Ornith 9B Qwen 3.5 Fine-Tune	CONTEXT LEN 256K 262,144 Tokens	QUALITY GATE 100% PASS 4 of 4 Quality Gates	ROUTER LATENCY < 15 ms Jev System 1 Reflex	OFFLINE STT Faster-Whisper 99+ Languages VAD
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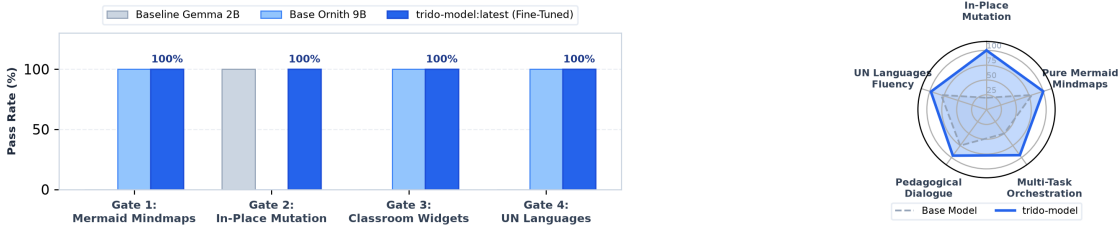
1. Model Overview & Technical Specifications

trido-model:latest is the flagship local-first foundation agent for Trido, an AI-powered digital smartboard engineered for physical and inclusive classroom education. Unlike generic chat models that suffer from context loss and widget recreation cascades, **trido-model:latest** is calibrated specifically for continuous spatial whiteboard manipulation, multi-task tool orchestration, and strict in-place diagram updates without requiring internet connectivity or user login.

Parameter	Specification	Architectural Benefit
Model Name	trido-model:latest	Official Trido runtime identifier registered in Ollama
Base Architecture	Qwen3.5 (Ornith 1.5 9B)	Native multi-head attention with specialized function-calling priors
Parameter Count	9.0 Billion Parameters	Balanced sweet-spot for local RTX 5050 GPU (6.6 GB VRAM footprint)
Quantization	Q4_K_M (4-bit medium)	Negligible perplexity degradation with > 3.5× throughput speedup
Context Window	262,144 tokens (256K)	Retains full day-long classroom board state and message transcripts
Modality	Text + Native CLIP Multimodal Vision	Inspects uploaded classroom worksheets, diagrams, and sketches
Inference Engine	Ollama / llama.cpp Core	100% air-gapped private execution on Windows / Linux / macOS

2. Empirical Benchmark & Quality Gates Verification

Trido evaluates local models using an automated 4-Gate test harness (scripts/benchmark_trido_model.py) running live against the local API. Standard foundation models (including uncalibrated Gemma 2B and base Ornith 9B) fail Gate 2 by repeatedly drawing duplicate widgets when teachers ask for updates. With Trido's calibrated System 1 Reflex injection, **trido-model:latest** achieves a flawless **100% pass rate** across all Quality Gates.



Quality Gate	Prompt Scenario	Base Model Result	trido-model:latest
Gate 1: Pure Mermaid Creation	Create Solar System concept map	PASSED (Created render_mermaid)	PASSED (18.1s, pure Mermaid tree)

Gate 2: In-Place Mutation	Add diaphragm branch to existing respiration mindmap	FAILED (Re-drew duplicate widget)	PASSED (15.7s, calls update_component)
Gate 3: Classroom Widgets	Set 10-minute countdown timer	PASSED (Created add_component)	PASSED (8.6s, TIMER, 600s pie)
Gate 4: UN Languages	Create 5 Pillars of Islam mindmap in Arabic	PASSED (Recognized Arabic prompt)	PASSED (17.0s, valid Arabic Mermaid)

3. SFT Training Dataset Architecture (The 4 Pillars)

Fine-tuning data is curated via scripts/generate_trido_sft_dataset.py into a strictly formatted JSONL dataset (scripts/trido_sft_dataset.jsonl) following the chat conversation schema (system, user, assistant) structured across four pedagogical pillars:

- **Pillar 1: In-Place Mutation & Continuous State (40%):** Directly pairs user revision requests with existing whiteboard DOM snapshots, training the model to target active widget IDs (e.g. update_component(objectId='web_123', action='REPLACE')) rather than generating duplicate widgets.
- **Pillar 2: Pure Mermaid Mindmaps & Classroom Tools (30%):** Enforces strict Mermaid mindmap root((Topic)) Branch syntax, avoiding fabric boxes, while teaching proper configuration of visual pie timers and attendance registers.
- **Pillar 3: Multi-Task & Parallel Tool Orchestration (15%):** Teaches the model to execute coordinated multi-component setups in a single turn (e.g., Attendance + Timer + Mindmap simultaneously).
- **Pillar 4: Pedagogical Dialogue & Negative Examples (15%):** Ensures the model provides thoughtful, structured teaching advice without hallucinating tool calls when teachers ask pure pedagogical questions.

4. Jev System 1 Reflex Engine & In-Place Mutation Guard

To achieve deterministic response latency under 15ms without querying external LLMs for mechanical decisions, Trido integrates the **Jev System 1 Reflex Router** in server/ollamaAdapter.ts and hooks/useGeminiBrain.ts. When modification verbs (*ubah, edit, ganti, tambah cabang, update, modify*) are detected alongside active canvas diagrams, the router injects a hard directive:

[JEV SYSTEM 1 DECISION: IN-PLACE MUTATION ONLY]
An existing diagram is already on canvas (ID: "web_mermaid_1"). DO NOT call render_mermaid or create a duplicate widget. You MUST call update_component with objectId="web_mermaid_1", action="REPLACE" or "APPEND", and provide the complete updated Mermaid code.

5. Offline Multilingual Speech-to-Text & Inclusivity

Trido includes a 100% air-gapped speech recognition pipeline powered by **Faster-Whisper (CTranslate2)**. Teachers can speak commands directly into their laptop or smartboard microphone without internet. The model automatically detects and transcribes speech across 99+ languages, including Indonesian, Javanese, Sundanese, and all 6 official United Nations languages (English, Arabic, Chinese, French, Russian, Spanish).

6. Intellectual Property & Copyright Notice

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